

RESEARCH OVERVIEW

My primary research interests lie in computational learning theory, with a particular focus on understanding the PAC-learnability of conditional (or selective) learning models and threshold-gate neural networks. My work also explores the learning of interpretable and fair representations, as well as other fundamental problems central to building reliable and trustworthy machine learning systems.

EDUCATION

Aug. 2019 – Aug. 2025 **Ph.D. in Computer Science**

Washington University in St. Louis, MO, USA

Thesis: Learning From Conditional Data Distributions, advised by Brendan Juba

Aug. 2016 – Dec. 2018 **M.S. in Electrical Engineering & Computer Science**

Washington University in St. Louis, MO, USA

Project: Learning Low-Energy Threshold Gate Networks, advised by Brendan Juba

Sept. 2011 – May 2015 **B.E. in Automation**

Beijing Jiaotong University, Beijing, CHINA

Project: Boosting Massive Image-Search System using FPGA, advised by Shikui Wei

RESEARCH EXPERIENCE

Aug. 2019 – Aug. 2025 **Washington University in St. Louis, MO, USA**

Research Assistant at *McKelvey School of Engineering*

Advisor: Brendan Juba

Learning From Conditional Data Distributions

- Studied a family of problems whose objective is to model properties of sub-populations of a distribution, typically by jointly learning a sub-population representation along with the target representation, under PAC-learning semantics.
- Developed a computationally efficient algorithm for *label-agnostic* linear classification conditioned on sub-populations represented by halfspaces under Gaussian distributions, with a provable approximation guarantee. (Joint work with Brendan Juba)
- Defined a family of *well-behaved* distributions that generalizes Gaussian and many other common distributions. Proved that our conditional classification algorithm may achieve comparable performance guarantees under this broader distribution class, and empirically evaluated the algorithm in PyTorch¹. (Joint work with Brendan Juba)
- Proved that approximating classification tasks conditioned on general halfspaces within a small additive error is intractable under standard cryptographic hardness assumptions. (Joint work with Brendan Juba)
- Proposed a *personalized prediction scheme* in which an individual predictor is learned for each incoming query, rather than a single universal predictor shared across all future queries. (Joint work with Brendan Juba)
- Developed an efficient algorithm for personalized prediction that jointly learns a halfspace containing the query point and a sparse linear classifier with provable performance guarantees on that halfspace under well-behaved distributions. The sparse classifier produced by our algorithm achieved competitive performance compared with more complex models in empirical evaluations², while offering superior interpretability due to its simplicity. (Joint work with Brendan Juba)
- Studied the problem of *auditing* subgroup fairness, which aims to determine whether a decision-making process discriminates against any demographic subpopulation.
- Established that the auditing problem remains computationally intractable when subpopulations are represented by membership functions as simple as halfspaces—even under Gaussian distributions—based on standard cryptographic hardness assumptions. (Joint work with Daniel Hsu and Brendan Juba)

¹<https://github.com/JizhouHuang/conditional-classification-with-homogeneous-selectors>

²<https://github.com/JizhouHuang/personalized-prediction-with-homogeneous-halfspaces>

- Reduced the auditing problem for homogeneous halfspace subgroups to learning homogeneous halfspaces under Gaussian distributions. Developed an auditing algorithm with provable guarantees by leveraging a Polynomial-Time Approximation Scheme for learning homogeneous halfspaces. (Joint work with Daniel Hsu and Brendan Juba)
- Developed an efficient improper auditing algorithm for subgroups represented by conjunctions, which is able to identify an unfairly treated subgroup defined by some polynomial threshold function. (Joint work with Daniel Hsu and Brendan Juba)

Aug. 2022 – May 2023 **UC Berkeley, CA, USA**

Visiting Scholar at *Simon's Institute*

- Participated in the *Data-Driven Decision Processes* and *Meta-Complexity* programs.
- Listened various talks of topics across multiple areas.

Oct. 2016 – May 2019 **Washington University in St. Louis, MO, USA**

Student Researcher at *Mckelvey School of Engineering*

Advisor: Brendan Juba

Learning Low-Energy Threshold Gate Networks

- Studied the PAC-learnability of sparsely activated (low-energy) threshold-gate networks. Investigated the relationship between the Unique Games Conjecture on expander graphs and such low-energy networks. (Joint work with Zihao Deng and Brendan Juba)
- Proposed an SVM-like program that captures different activation patterns of the gates, enabling the application of multiple robust clustering methods. Proved an energy upper bound on the number of variables involved in each constraint, which significantly reduced the program's complexity. (Joint work with Zihao Deng and Brendan Juba)
- Showed that the set of possible weights for each feasible activation pattern forms a convex cone over the hinge loss, indicating that low-energy networks with different hinge losses can be distinguished by the angle between their vectorized parameters. (Joint work with Zihao Deng and Brendan Juba)
- Formulated the problem as a spherical optimization task that fits within the geodesic optimization framework, allowing strong convexity to be incorporated into the loss function. (Joint work with Zihao Deng and Brendan Juba)

Dec. 2014 – Mar. 2016 **Beijing Jiaotong University, Beijing, CHINA**

Student Researcher at *Electronic Information Engineering Institute*

Advisor: Shikui Wei

Boosting Massive Image-Search System using FPGA

- Built a *k-Nearest Neighbor* based image-searching system on PC using C#, and an image-searching system of the same algorithm on FPGA using C++.
- Applied optimization instructions to the C++ code using Vivado High-Level Synthesis Software, which significantly improved the throughput of the algorithm on FPGA. Integrated the optimized program into a Zedboard embedded system using Vivado Software and set up multiple properties of the system using C on Xilinx SDK.
- Configured AXI Direct Memory Access via Xilinx SDK to achieve high-speed data exchanging between an on-board SD card and Xilinx Artix-7 FPGA, which solved the problems caused by memory deficiency and slow data transfer.

TEACHING EXPERIENCE

Spring 2024 **Washington University in St. Louis, MO, USA**

Teaching Assistant at *Mckelvey School of Engineering*

CSE 513T Theory of Artificial Intelligence and Machine Learning

- Filled in for multiple lectures.

Fall 2021 **Washington University in St. Louis, MO, USA**

Teaching Assistant at *Mckelvey School of Engineering*

CSE 513T Theory of Artificial Intelligence and Machine Learning

- Filled in for multiple lectures.
- Graded students' homework and held weekly TA sessions to answer students' questions.

PUBLICATIONS

- D. Hsu, **J. Huang**, and B. Juba. Distribution-Specific Auditing For Subgroup Fairness. *The 5th annual Symposium on Foundations of Responsible Computing*, Cambridge, MA, USA, 2024.
- **J. Huang**, B. Juba. Distribution-Specific Agnostic Conditional Classification With Halfspaces. *The 13th International Conference on Learning Representations*, Singapore, 2025.
- **J. Huang**, B. Juba. Personalized Prediction By Learning Halfspace Reference Classes Under Well-Behaved Distribution. Preprint, arXiv: <https://arxiv.org/abs/2509.15592>.

TALKS

- “Distribution-Specific Auditing For Subgroup Fairness” in *The 5th annual Symposium on Foundations of Responsible Computing*, Harvard University, Cambridge, MA, June 13, 2024.
- “Subgroup-based Fairness Auditing - from a computational complexity view” in *Doctoral Student Research Seminar(DSS)*, Washington University in St. Louis, St. Louis, MO, November 17, 2023.
- “An Efficient Algorithm For Fairness Auditing” in *Doctoral Student Research Seminar(DSS)*, Washington University in St. Louis, St. Louis, MO, October 22, 2021.

AWARDS

- FORC 2024 Best Student Paper

POSTERS

- “Distribution-Specific Agnostic Conditional Classification With Halfspaces” in *The 13th International Conference on Learning Representations*, Singapore, April 24th, 2025.
- “Auditing Subgroup Fairness on Nice Distributions” in *NSF Fairness in AI PI Meeting*, Arlington, VA, USA, 2023.

WORK EXPERIENCE

June 2025 – Present **Meta**, CA, USA

Research Scientist at *Connected Feed Modeling*

Manager: Junfeng Pan

- Migrated Facebook app Story In Feed module from old PyPer framework to new MVAI framework.
- Proposed ViSTA user history sequence generator to the main Connect Feed prediction model and improve the offline NE by 0.1%.(Team work)

May 2023 – Aug. 2023 **Meta**, CA, USA

Software Engineer Intern at *MRS Reels Recommendation*

Mentor: Na Zhang

Path-Based Trigger Network

- Created an attention-based two-tower network to model user-item similarity through user history for Facebook reels recommendations.
- Improved the two-tower model using Gram-Schmidt decomposition mechanism, resulting in reduced loss.
- Orchestrated Quality Enhancement (QE) experiments for both models and refined them based on real-world user feedback.

May 2022 – Aug. 2022 **Meta**, CA, USA

Software Engineer Intern at *Core ML*

Mentor: Wei Chang

Multi-interest Retrieval

- Worked on building multi-head two-tower network models to capture different human interests at retrieval stage for Facebook reels recommendation.
- Wrote a Presto script to analyze the pair-wise correlations between individual network models.
- Built and evaluated other competitive models, such as attention networks.